

Zoya Ejaz

Curious problem-solver driven to turn complex challenges into practical, impactful solutions.

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EDUCATION

Vellore Institute of Technology

Jun, 2023 - May, 2027

B.Tech in Computer Science & Engineering (Artificial Intelligence and Machine Learning)

CGPA: 8.79 / 10.00

SKILLS

- **Languages:** Python, Java, SQL
- **Machine Learning:** Scikit-Learn, Ensemble Learning, Feature Engineering, PCA, ANOVA, Regression, Classification, Clustering, Association Rule mining
- **Deep Learning & Computer Vision:** CNNs, OpenCV, RNN, LSTM, GRU
- **NLP:** TF-IDF, Text Classification, Prompt Engineering, LLM Fundamentals
- **Data:** Pandas, NumPy, Matplotlib, Seaborn, EDA, Data Visualization, Excel, Streamlit
- **Generative AI & Tools:** LLMs, Prompt Engineering, RAG, LangChain, Docker, Git

PROJECTS

JobShield AI — Recruitment Scam Detection System

Jan 2026 - March 2026

Tech Stack: Python, Pandas, NumPy, Scikit-Learn, NLP, TF-IDF

[Github](#)

- Addressed the growing risk of fake job postings and recruitment scams targeting job seekers in India.
- Built an ML-based system to identify suspicious job listings and assess their legitimacy.
- Applied NLP, TF-IDF, machine learning, rule-based features, and network analysis to detect scam patterns.
- Built an explainable risk-scoring system to detect fraudulent jobs and suspicious recruiter networks.

Glaucoma Disease Detection and Classification

April 2026 - June 2026

Tech Stack: Python, Scikit-Learn, OpenCV, Scikit-Image, NumPy, SciPy, Pandas

[Github](#)

- Engineered an automated computer vision disease detection system using IEMD and ensemble learning.
- Extracted 504+ multi-descriptor features and optimized model dimensionality using ANOVA and PCA.
- Quadrupled dataset size to 2,020 images via geometric augmentation to effectively prevent overfitting.
- Published an **IEEE** paper presenting a predictive model achieving 94.85% specificity.

NLP2SQL, Natural Language to SQL Query Generation System

Sept 2026 - Present

Tech Stack: Python, LangChain, Groq API, SQLAlchemy, sqlglot, Streamlit

[Github](#)

- Built a two-stage LLM pipeline for natural language to validated SQL generation.
- Implemented AST-based SQL validation using sqlglot to prevent destructive queries and SQL injection.
- Designed live schema introspection and self-correcting LLM retries using execution errors.
- Developed an evaluation harness and Streamlit dashboard to benchmark accuracy and demo queries live.

CERTIFICATIONS

- **University of Michigan (Coursera):** Certified in Applied Predictive Modeling and Feature Engineering.
- **Deloitte:** Completed Data Analytics simulation involving visualization and business insights.
- **AWS (Forge):** Earned a certificate for completing the Solutions Architecture Job Simulation.

ACHIEVEMENTS & LEADERSHIP

- **Event Lead, Eureka Club:** Orchestrated large-scale tech events while managing a 10-member team.
- **Placement Cell Member:** Assisted in Cognizant recruitment coordination for a cohort of 300+ students.
- **Smart India Hackathon 2024:** Selected as a College Finalist among competing teams for developing an innovative solution to a real-world problem.